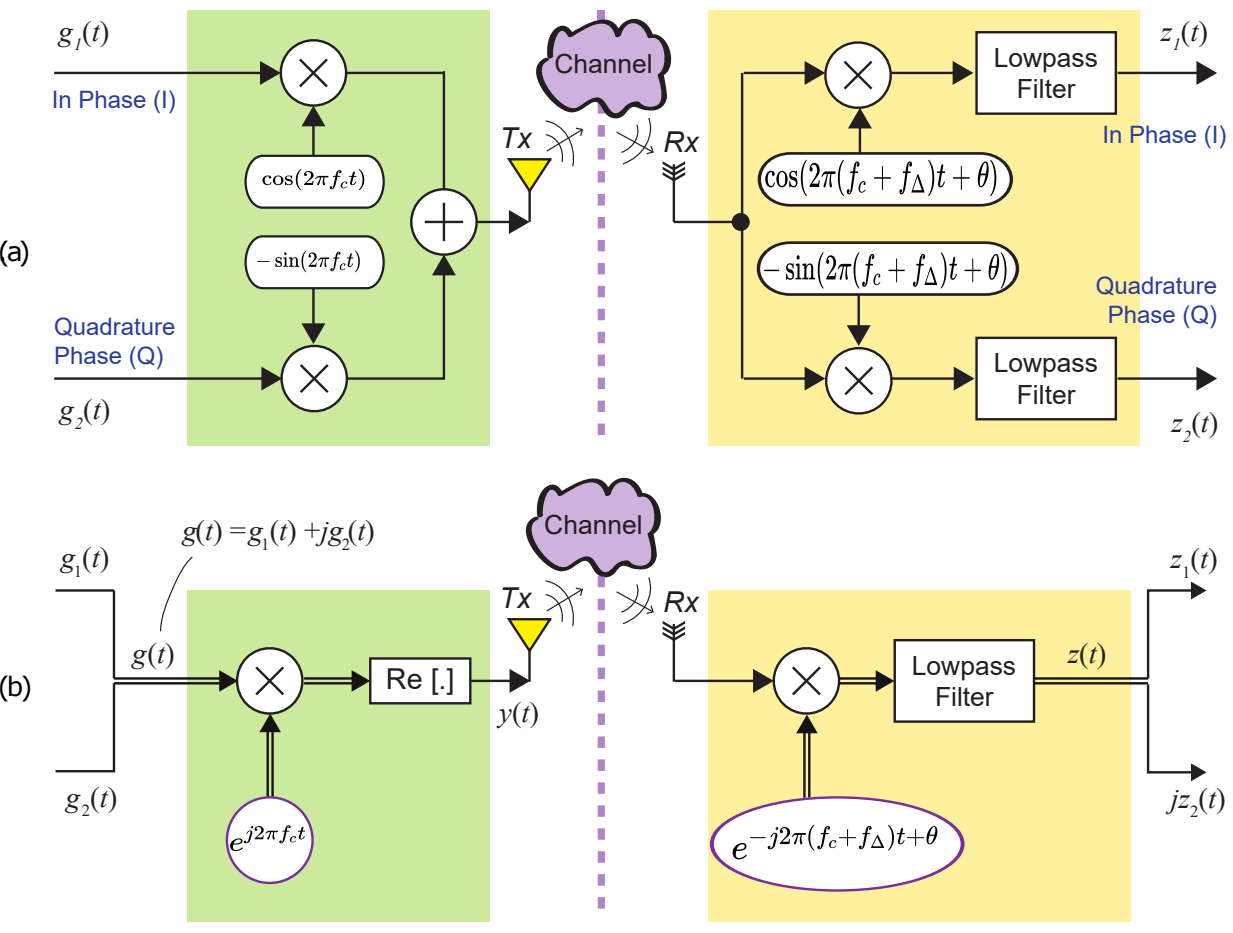

(a)
$g_1(t)$
In Phase (I)
$\cos(2\pi f_c t)$
$-\sin(2\pi f_c t)$
Quadrature Phase (Q)
$g_2(t)$
Channel
Tx
Rx
$\cos(2\pi(f_c+f_\Delta)t+\theta)$
$-\sin(2\pi(f_c+f_\Delta)t+\theta)$
Lowpass Filter
Lowpass Filter
$z_1(t)$
In Phase (I)
Quadrature Phase (Q)
$z_2(t)$
(b)
$g(t)=g_1(t)+jg_2(t)$
$g_1(t)$
$g(t)$
$g_2(t)$
Re [.]
$e^{j2\pi f_c t}$
Channel
Tx
Rx
$y(t)$
Lowpass Filter
$z(t)$
$e^{-j2\pi(f_c+f_\Delta)t+\theta}$
$z_1(t)$
$jz_2(t)$